

DIGITAL FORENSICS

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References

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 - Bill Nelson, et al. Cengage, 2025
- Digital Forensics and Incident Response, 4th Edition,
 - Gerard Johansen, Packt, 2025
- Roadmap of digital forensics investigation process with discovery of tools, Cyber Security and Digital Forensics,
 - Anita Patil, et al. (2022):241-269.
- SWGDE 18-F-001-2.0 / SWGDE 18-F-002-2.0 /
- SWGDE 17-F-002-2.1 / SWGDE 23-Q-001-1.1 /
- NIST SP 800-61 Rev. 3
- IETF RFC 3227
- <https://www.sans.org/blog>

DAY 4

EXAMINATION AND ANALYSIS OF THE DIGITAL EVIDENCE & REPORTING

WINDOWS



<https://www.sans.org/posters/hunt-evil>

- Knowing what's normal on a Windows host helps cut through the noise to quickly locate potential malware.
- Use the poster as a reference to know what's normal in Windows and to focus your attention on the outliers.

WINDOWS FILESYSTEM

NTFS

NTFS – Windows File system

- On an NTFS disk
 - First data set is the **Partition Boot Sector**
 - Next is **Master File Table (MFT)**
 - MFT contains information about all files on the disk
 - Including the system files the OS uses
 - Records in the MFT are called **metadata**

NTFS Timestamps – MACB

NTFS commonly stores four timestamps, often called **MACB**

- **M**odified
- **A**ccessed
- **C**hanged (\$MFT modified)
- **B**irth (file creation)



Available in SIFT Desktop

NTFS Timestamps – MACB

- NTFS file timestamps are commonly stored in two attributes within the file's \$MFT record:
 - \$STANDARD_INFORMATION (\$SI)
 - \$FILE_NAME (\$FN)

NTFS Timestamps – \$SI

- \$STANDARD_INFORMATION (\$SI)
 - Contains the primary MACB
 - Updated during normal file activity (user level processes).
 - Can be modified through Windows APIs and timestomping tools.
 - Therefore, it may be less reliable when **timestamp manipulation** is suspected.

NTFS Timestamps – \$FN

- \$FILE_NAME (\$FN)
 - Contains the filename, another set of MACB timestamps, ...
 - Can only be modified by the system kernel
 - Primarily updated during file-system namespace operations, such as creating, renaming, or moving a file.
 - More difficult to modify using normal user-level tools.

Timestomping

- Timestomping is an anti-forensics technique used to modify a file's timestamps to hide or misrepresent when the file was created, modified, accessed, or changed.
- Investigators detect possible timestomping by
 - Comparing \$SI and \$FN timestamps
 - Mismatch may indicate manipulation

MACB Extraction From \$MFT

In Windows:

```
MFTECmd.exe -f 'E:\$MFT' --body E:\output --bodyf mft.body --blf --bd1 C:\
```

In SIFT:

```
mactime -z UTC -y -d -b mft.body > file-system_timeline.csv
```

AUTOPSY TIMELINE ANALYSIS

Timeline - Editor

Timeline

Display Times In: ☒ Local Time Zone ☐ GMT / UTC

History Back Forward

Filters

☒ Apply ☐ Default

☐ Text Filter

enter filter string

☐ Tagged Items

☐ Hash Sets

☒ Data Source

☐ File Types

☒ Event Type

☒ File System

☒ Web Activity

☒ Misc Types

☒ Custom Types

View Mode:

Counts

Details

List

+ Add Event

Snapshot Report

Refresh View

37,382 events

Date/Time	Event Type	Description	Tagged	Hash Hit	
2012-03-22 19:30:16	MA_C	/Documents and Settings/Peter/Local Settings/Application Data/Microsoft/Windows/UsrClass.dat.LOG			
2012-03-22 19:30:16	M_BC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/0/3A			
2012-03-22 19:30:16	MABC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/5/80/B0E15d01			
2012-03-22 19:30:16	MABC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/D/50/71C4Ed01			
2012-03-22 19:30:16	Web History	http://www.citizensbank.com/promotions/brand/sem/bran ... 8529537&005=359290367&006=8358382937&007=Search&008=			
2012-03-22 19:30:16	Web Cookies	www.citizensbank.com			
2012-03-22 19:30:16	M_BC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/D/4C			
2012-03-22 19:30:16	MABC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/D/4C/EC63Bd01			
2012-03-22 19:30:16	M_BC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/D/50			
2012-03-22 19:30:16	MABC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/0/3A/338F0d01			
2012-03-22 19:30:16	___C	/Documents and Settings/Peter/Local Settings/Application Data/Microsoft/Windows/UsrClass.dat			
2012-03-22 19:30:16	M_BC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/1/B6			
2012-03-22 19:30:16	MABC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/1/B6/FB29Ed01			
2012-03-22 19:30:16	MABC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/F/F0/4B6E8d01			
2012-03-22 19:30:16	M_BC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/5/80			
2012-03-22 19:30:16	M_BC	/Documents and Settings/Peter/Local Settings/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/Cache/F/F0			
2012-03-22 19:30:16	Web History	http://www.citizensbank.com/promotions/brand/sem/bran ... 8529537&005=359290367&006=8358382937&007=Search&008=			
2012-03-22 19:30:17	Web Cookies	www.citizensbank.com			
2012-03-22 19:30:17	M_BC	/Documents and Settings/Peter/Application Data/Mozilla/Firefox/Profiles/0ozf9kh7.default/downloads.solite			

Start: Aug 27, 2011 4:45:25 AM

Jump By: Year

End: Sep 23, 2017 10:28:48 AM

Hidden Descriptions

Hex

Text

Application

Message

File Metadata

Results

Annotations

Other Occurrences

Download Images

CitizensBank.com | Login to Online Banking | Locate a Branch/ATM

Need Help? Call Citizens Bank

We treat customers like neighbors. Because they usually are.

Whatever you need, we're here to help you find the right solution.

• Personal

• Small Business

WINDOWS EVENTS/LOGS

Windows Events

- Security Log
 - Logon successes and failures: 4624, 4625
 - Special privileges assigned: 4672
 - Account creation or modification: 4720, 4722, 4724, 4732
 - Process creation: 4688, when auditing is enabled
 - Log clearing: 1102

Windows Events

- Security Log
 - Scheduled task created: 4698
 - Scheduled task deleted: 4699
 - Scheduled task updated: 4702

Windows Events References

- <https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/plan/appendix-l--events-to-monitor>
- <https://www.sans.org/posters/hunt-evil>

NETWORK FORENSICS

Sources of Network Evidence

- Routers/switches logs
- Web proxy logs
- Firewall
- DHCP/DNS/Web servers logs
- Packet capture (full pcap)
- NetFlow
 - Statistical record of packets with common attributes / key values
 - No payload

Network Forensics – Forensic Value

- Network Forensics is the process of:
 - Capturing and analyzing network traffic, with the objective of information gathering, incident identification
 - To trace the origin of attacks.
 - To detect and understand malicious activities.
 - To collect evidence for legal proceedings.

Network Forensics – Where?

- Network logs record ingoing and outgoing traffic
 - Network servers
 - For example, proxy server logs
 - Routers
 - Firewalls
- Tcpdump and Wireshark - tools for examining network traffic
 - Can generate top 10 lists
 - Can identify patterns

Network Forensics – investigating proxy logs

```
(kali㉿kali)-[~]  
$ sudo tail -f /var/log/squid/access.log  
192.168.8.162 - - [02/Apr/2022:19:50:47 -0400] "CONNECT firefox.settings.services.mozilla.com:443 HTTP/1.1" 200 7669  
"- "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:98.0) Gecko/20100101 Firefox/98.0" TCP_TUNNEL:HIER_DIRECT
```

Annotations:

- Client IP address (who requested this website)
- Timestamp
- HTTP Method
- The requested website (URI)
- HTTPS
- (user-agent)
The browser name and version
- HTTP Status Code

Network Forensics – investigating proxy logs

```
(kali㉿kali)-[~]  
$ sudo grep -v " CONNECT " /var/log/squid/access.log | awk '{print $7}' | sort |  
uniq -c | sort -nr | head -n 20  
grep: /var/log/squid/access.log: binary file matches
```

```
90 cdn.syndication.twimg.com:443  
71 uj.edu.sa:443  
64 img-s-msn-com.akamaized.net:443  
64 http://ocsp.digicert.com/  
58 ib.adnxs.com:443  
51 login.microsoftonline.com:443  
48 www.google.com:443  
36 www.uj.edu.sa:443  
33 http://ocsp.pki.goog/gts1c3  
32 www.moi.gov.sa:443  
32 static-global-s-msn-com.akamaized.net:443  
31 www.absher.sa:443  
31 safebrowsing.googleapis.com:443  
25 teams.events.data.microsoft.com:443  
25 ksu.edu.sa:443  
24 securepubads.g.doubleclick.net:443
```

**How many times each website has been requested.
Ordered by the most frequent websites**

Network Forensics – investigating proxy logs

```
(kali㉿kali)-[~]  
$ sudo grep -v "CONNECT" /var/log/squid/access.log | awk '{print $1,$4,$7}' |  
grep pastebin.com | sort  
grep: /var/log/squid/access.log: binary file matches  
192.168.8.162 [02/Apr/2022:19:36:36 www.pastebin.com:443  
192.168.8.162 [02/Apr/2022:19:45:22 pastebin.com:443  
192.168.8.162 [02/Apr/2022:19:49:39 pastebin.com:443
```


Network Forensics – investigating proxy logs

```
(kali㉿kali)-[~]  
$ sudo grep "\"GET" /var/log/squid/access.log | awk '{print $7}' | sort | uniq -c |  
sort -nr | head  
grep: /var/log/squid/access.log: binary file matches  
33 http://edgedl.me.gvt1.com/edgedl/release2/chrome_component/adfnozgc6k2uqmgswpb  
jjuv577ga_20220323.437574594/obedbbhbpmojnkanicioggnmelmoomoc_20220323.437574594_all_E  
NUS500000_advueet2dqbrz7uyng25kkjp5p5a.crx3  
16 http://edgedl.me.gvt1.com/edgedl/release2/chrome_component/aotje56gqr7oar63brt  
hh5wrmq_99.280.200/gkmgaoaipdjhmangpemjhigmamcehddo_99.280.200_win64_bliouwmrme4laeka  
3ebtf7veq.crx3  
9 http://ctldl.windowsupdate.com/msdownload/update/v3/static/trustedr/en/DF3C24F  
9BFD666761B268073FE06D1CC8D4F82A4.crt?  
8 http://edgedl.me.gvt1.com/edgedl/release2/chrome_component/adiemnuog25sdcd4jwc  
qtw2ewfza_102.0.4980.0/jamhcnkihinmdlkakkaopbjbbcngflc_102.0.4980.0_all_adadww2d4c7pp  
g7axpwekwd7xwka.crx3  
5 http://edgedl.me.gvt1.com/edgedl/release2/chrome_component/APhHMzuprJvS7ixvnAk  
_gdI_1/anGnv31dm0JhheXBnYQ3gw  
5 http://edgedl.me.gvt1.com/edgedl/release2/chrome_component/acxolis3zo2arxlttyf  
polzxixxa_27.8/imefjhfbkmcmebodilednhmaccmincoa_27.8_win_e74raku26b5hr257m3sgtyxbpm.c  
rx3  
4 http://edgedl.me.gvt1.com/edgedl/release2/chrome_component/ad6ioqohe4v2uzgmdk4  
cg6iwyq6q_2022.3.28.1141/ggkkehgbnfjpeggfpleeakpidbkibbm_n_2022.3.28.1141_all_ad2642qic  
jxpnvhlrxrsgp77tika.crx3  
4 http://cacerts.digicert.com/DigiCertGlobalRootG2.crt  
3 http://uj.edu.sa/  
2 http://um.edu.sa/sites/default/files/styles/slider_image/public/2021-12/%D9%85  
%D9%88%D9%82%D8%B91.png?
```

Cloud Forensics

- Evidence may come from:
 - virtual machines,
 - cloud storage,
 - audit logs,
 - identity services,
 - databases, and
 - network logs

Cloud Forensics

- It supports investigations involving account compromise, data breaches, insider threats, malware, and unauthorized access.
- **Key challenges** include shared infrastructure, limited physical access, data location, provider dependency, encryption, and legal jurisdiction.
- Investigators often rely on cloud-provider logs, snapshots, API records, and access histories rather than direct access to physical devices.



Reporting

Autopsy to Generate Reports

- Exports forensic findings from a case into a structured report.
- Investigators can include all analysis results or only tagged/relevant items.
- Reports may contain recovered artifacts, keyword hits, file details, hashes, and other examination results.
- Supported formats include HTML, Excel, KML, tab-delimited files, and timeline body files, depending on the installed reporting modules.
- Generated reports are stored in the case's Reports section and can be reviewed or shared with other investigators.

Report Navigation

-  Case Summary
-  Accounts: Device (10)
-  Accounts: Email (10)
-  Accounts: Phone (80)
-  Accounts: Words with Friends (5)
-  Call Logs (216)
-  Contacts (24)
-  Extension Mismatch Detected (1)
-  GPS Route (9)
-  GPS Trackpoints (1)
-  Hashset Hits (2)
-  Interesting Results (3)
-  Keyword Hits (367)

UNCLASSIFIED

Autopsy Forensic Report

HTML Report Generated on 2018/12/17 09:31:34

Case:	Case 7
Case Number:	No case number
Examiner:	John Doe
Number of Images:	6

Image Information:

image1.vhd

Timezone:	America/New_York
Path:	R:\work\images\image1.vhd

image3.vhd

Reporting Your Findings

- Details to include in your report
 - Report title (e.g., examination report, amended report)
 - Case identifier and requester
 - Clearly stated purpose of the examination
 - Description of the processes and results of tests and examinations with supporting data, as needed (e.g., automated tool reports, screen captures and other exhibits)
 - Explanation of conclusions/opinions drawn from the data, if appropriate
 - Disposition of evidence
 - Report authorization

<https://www.swgde.org/documents/published-complete-listing/18-q-002-swgde-requirements-for-report-writing-in-digital-and-multimedia-forensics/>

Lab 4: Examination and Analysis of The Digital Evidence

Part 2